

SOFTWARE TESTING FUNDAMENTALS

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS13	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

To learn Software Testing has different goals and objectives:

1. Finding defects which may get created by the programmer while developing the software.
2. Gaining **confidence** in and providing information about the level of quality.
3. To prevent defects.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to:

1. Understand software testing methods
2. apply various software testing techniques
3. Design and conduct a software test process for a software testing project
4. Designing solutions for various software testing problems by selecting appropriate software test model
5. Implement various practice oriented software testing projects

UNIT - I	INTRODUCTION	CLASSES: 08
Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.		
UNIT - II	TESTING METHODOLOGIES	CLASSES: 12
White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.		
UNIT - III	INTEGRATION TESTING	CLASSES: 10
Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing		
UNIT - IV	TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING	CLASSES: 10
Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.		
UNIT - V	TEST MANAGEMENT AND AUTOMATION TEST PLANNING	CLASSES: 10
Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool framework, Test tool selection.		

TEXT BOOKS
1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education. 2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.
REFERENCE BOOKS
1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley 2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.